

Filip Skogh

📍 Zürich, Switzerland

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PROFILE

3+ years of experience in Python and C++. Hackathon enjoyer with over CHF 5000 in prize money. Enthusiastic about writing efficient code related to LLMs and ML. Research experience with Transformers (both encoder and decoder) at IBM Research and video segmentation at ETH Zürich. Knowledge in local inference using `llama.cpp`.

EDUCATION

ETH Zürich SEMP student, M.Sc. Computer Science

Zürich, Switzerland
Sep 2022 - Sep 2023

- **Activities:** ETH Analytics Club
- **Courses:** Natural Language Processing, Advanced Machine Learning, Computer Vision Project

Chalmers University of Technology M.Sc. Data Science and AI

Gothenburg, Sweden
Aug 2021 - Sep 2023

- **GPA:** 4.8/5.0
- **Courses:** Non-linear Optimization, Stochastic Processes and Bayesian Inference, Algorithms Computer Vision, Reinforcement Learning, Image Analysis, Large-scale Distributed Computation

Nanyang Technological University Exchange Student, B.Sc. Computer Science

Singapore
Jan-Jun 2020

- **Courses:** Digital Signal Processing, Cryptography, Operating Systems, Computer Networks

Luleå University of Technology B.Sc. Computer Science and Engineering

Luleå, Sweden
Aug 2018 - Jun 2021

- **GPA:** 5.0/5.0

EXPERIENCE

Research Intern IBM Research

Zürich, Switzerland
May-Current 2024

- Meta-learning with hyper-networks, researching encoder-decoder networks that predict weights for other networks.
- ICCV paper in review proposing a new optimal transport normalization layer for Vision Transformers.
- Full research cycle with ideation, literature review, experimentation, and peer-review process.

Junior ML Engineer Logiblox

Zürich, Switzerland
Oct 2023 - Apr 2024

- Increased token generation speed by over 300% for LLM inference with quantization and caching.
- Built compiler that converts visual no-code representation in to executable python code.
- Set up from scratch a dockerized server for inference with CI/CD pipeline.

Master Thesis Computer Vision Lab ETH Zürich

Zürich, Switzerland
Feb-Sep 2023

- Used spatial and temporal features to construct a self-supervised loss function for video segmentation.
- Multi-GPU training on 8xA6000 node with DDP on 100+ GB video datasets.
- Supervised by Prof. Fisher Yu and Dr. Martin Danelljan. filipskogh.com/thesis.pdf

Research Intern University of Massachusetts Amherst

Massachusetts, United States
Jun-Sep 2022

- Developed a request scheduler for data-centers that minimize carbon emissions.
- Demonstrated a 70% reduction in carbon emissions without latency degradation.
- Best Paper Runner-up award at the IEEE IGSC 2023 conference in Toronto, Canada.

Software Intern Orange Cyberdefense

Stockholm, Sweden
Summers 2019 - 2021

- Developed threat response software in Python that block ransomware, C&C servers and take snapshots for forensics. Deployed globally on 50.000+ end points in 70+ countries.

AWARDS

Lab 42 Hackathon at the World Economic Forum

Davos, Switzerland
Jan 2025

- Won first prize (CHF 5000) as part of a 3-person team.
- Developed an interactive avalanche segmenter using SAM 2.1, also trained a ResNet model to classify the avalanche type. Predicted the avalanche size using geometric computer vision. All wrapped behind FastAPI with a react front-end.

PROJECTS

Streaming JSON parser 🐞: Implemented a C++JSON parser with streaming support at the character level suited for auto-regressive LLMs. Exposed through a clean Python interface using Python bindings.

Packet intercept proxy: C++ project developed continuously for three years by hooking Windows socket API `send` and `recv` using the Detours C++library. Large-scale reverse engineering of project with 100k+ loc. Reverse engineered encryption protocols and ciphers to intercept traffic at packet level.

Packet fuzzer: Wrote a custom fuzzer in C++ to search op-code ranges and packet payload data to find server-side exploits for an MMORPG game.

Teaching Transformers arithmetic: Trained a GPT-style Transformer in PyTorch to learn addition with different tokenizers. Showed that if the digits are tokenized in reverse order the problem is easier.

Neural network certification 🐞: Developed custom network layers in PyTorch to propagate intervals through a network allowing us to deterministically prove properties about robustness, certain fairness guarantees and that adversarial attacks are not possible. This project was part of the course Reliable and Trustworthy AI at ETH.

Blockchain implementation: Implemented parts of the Bitcoin protocol from scratch to create, (i) a wallet address derived from an elliptic curve public key, (ii) a signed transaction which can be broadcasted to the network, and (iii) a block verifier.

Google Developer Student Club: Built an interactive 3D learning game in Unity3D C# by building a hospital simulator. The project idea was conceived by medical professors prompted by the pandemic and was aimed to simulate medical students' practicum. During the project I worked in close contact with medical professionals and translated medical procedures into implementable scenarios in-game.

RANSAC 🐞: Iterative parameter estimation using RANSAC with optimal hypothesis testing that minimizes the number of tests performed. The project was motivated by the scarcity of available implementation and was based on the original white paper.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, bash, Matlab, English, Swedish

Frameworks: llama.cpp, PyTorch, AFL++, Hugging Face, Map-Reduce, PySpark, OpenCV, Flask, Java Native Interface

Miscellaneous: Vim, Slurm, FastAPI, Docker, Git, Regex